A FINITE TYPE MODULE THAT IS FLAT AND PURE ALONG A FIBRE IS UNIVERSALLY PURE ALONG THAT FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Purity along a fibre and its behaviour under an elementary etale neighbourhood and under flat base change. Same chapter flat.tex, definition labelled definition-pure and the discussion following it (proof lines 5294-5295), and label lemma-flat-descend-pure (proof line 5303). Keeps purity after pulling back along an elementary etale neighbourhood, and lets purity of the base change along a flat pointed morphism descend to purity of the original, which is what makes the reduction in the preliminary paragraph legitimate.
- Existence of an affine etale neighbourhood on which the module has free global sections, and purity of such a module. Same chapter flat.tex, labels lemma-finite-type-flat-along-fibre-free-variant (proof line 5322) and lemma-affine-locally-projective-pure (proof line 5339). Produces the etale g : X' -> X with X' affine, X_s contained in g(X') and the global sections of the pullback free over the localised base, and gives that the pulled back module is universally pure over S.
- The purity criterion through relative associated primes and their base change behaviour. Same chapter flat.tex, label lemma-criterion (proof lines 5343 and 5357); Divisors, label divisors-lemma-base-change-relative-assassin (proof line 5347) and remark divisors-remark-base-change-relative-assassin (proof line 5348). Turns purity along the fibre into the inclusion of the relative associated primes of F over S inside the image of the etale morphism, and shows that this inclusion is stable under arbitrary base change of the pointed base, which yields universal purity.

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Source reference. Original lemma 05J4: sources/stacks-flat/pinned-original-source.tex, source lines 4808–4808. Navigation pointer for \ref{definition-pure} used inside the counted span (line 5295); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DC: sources/stacks-chow/tags.txt, source lines 6950–6950. Navigation pointer for \ref{divisors-lemma-base-change-relative-assassin} used inside the counted span (line 5347); the argument that consumes it is retained in the span.

Source reference. Original lemma 05KL: sources/stacks-chow/tags.txt, source lines 7205–7205. Navigation pointer for \ref{divisors-remark-base-change-relative-assassin} used inside the counted span (line 5348); the argument that consumes it is retained in the span.

Source reference. Original lemma 05K6: sources/stacks-flat/pinned-original-source.tex, source lines 5148–5148. Navigation pointer for \ref{lemma-affine-locally-projective-pure} used inside the counted span (line 5339); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L4: sources/stacks-flat/pinned-original-source.tex, source lines 5197–5197. Navigation pointer for \ref{lemma-criterion} used inside the counted span (line 5343); the argument that consumes it is retained in the span.

Source reference. Original lemma 05L1: sources/stacks-flat/pinned-original-source.tex, source lines 3627–3627. Navigation pointer for \ref{lemma-finite-type-flat-along-fibre-free-variant} used inside the counted span (line 5322); the argument that consumes it is retained in the span.

Source reference. Original lemma 05J9: sources/stacks-flat/pinned-original-source.tex, source lines 4949–4949. Navigation pointer for \ref{lemma-flat-descend-pure} used inside the counted span (line 5303); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$. Assume*

(1) *f is of finite type,*
(2) *$\mathcal{F}$ is of finite type,*
(3) *$\mathcal{F}$ is flat over S at all points of X_s, and*
(4) *$\mathcal{F}$ is pure along X_s.*

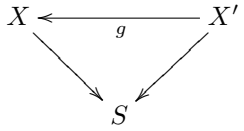
Then $\mathcal{F}$ is universally pure along X_s.

Proof. We first make a preliminary remark. Suppose that $(S', s') \to (S, s)$ is an elementary étale neighbourhood. Denote $\mathcal{F}'$ the pullback of $\mathcal{F}$ to $X' = X \times_S S'$. By the discussion following Definition 05J4 we see that $\mathcal{F}'$ is pure along $X'_{s'}$. Moreover, $\mathcal{F}'$ is flat over S' along $X'_{s'}$. Then it suffices to prove that $\mathcal{F}'$ is universally pure along $X'_{s'}$. Namely, given any morphism $(T, t) \to (S, s)$ of pointed schemes the fibre product $(T', t') = (T \times_S S', (t, s'))$ is flat over (T, t) and hence if $\mathcal{F}_{T'}$ is pure along $X_{t'}$ then $\mathcal{F}_T$ is pure along X_t by Lemma 05J9. Thus during the proof we may always replace (s, S) by an elementary étale neighbourhood. We may also replace S by $\mathrm{Spec}(\mathcal{O}_{S,s})$ due to the local nature of the problem.

Choose an elementary étale neighbourhood $(S', s') \to (S, s)$ and a commutative diagram

$$\begin{array}{ccc} X & \xleftarrow{g} & X' \\ \downarrow & & \downarrow \\ S & \longleftarrow & \mathrm{Spec}(\mathcal{O}_{S',s'}) \end{array}$$

such that $X' \to X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ is étale, $X_s = g((X')_{s'})$, the scheme X' is affine, and such that $\Gamma(X', g^*\mathcal{F})$ is a free $\mathcal{O}_{S',s'}$-module, see Lemma 05L1. Note that $X' \to \mathrm{Spec}(\mathcal{O}_{S',s'})$ is of finite type (as a quasi-compact morphism which is the composition of an étale morphism and the base change of a finite type morphism). By our preliminary remarks in the first paragraph of the proof we may replace S by $\mathrm{Spec}(\mathcal{O}_{S',s'})$. Hence we may assume there exists a commutative diagram



of schemes of finite type over S, where g is étale, $X_s \subset g(X')$, with S local with closed point s, with X' affine, and with $\Gamma(X', g^*\mathcal{F})$ a free $\Gamma(S, \mathcal{O}_S)$-module. Note that in this case $g^*\mathcal{F}$ is universally pure over S, see Lemma 05K6.

In this situation we apply Lemma 05L4 to deduce that $\mathrm{Ass}_{X/S}(\mathcal{F}) \subset g(X')$ from our assumption that $\mathcal{F}$ is pure along X_s and flat over S along X_s. By Divisors, Lemma 05DC and Remark 05KL we see that for any morphism of pointed schemes $(T, t) \to (S, s)$ we have

$$\mathrm{Ass}_{X_T/T}(\mathcal{F}_T) \subset (X_T \to X)^{-1}(\mathrm{Ass}_{X/S}(\mathcal{F})) \subset g(X') \times_S T = g_T(X'_T).$$

Hence by Lemma 05L4 applied to the base change of our displayed diagram to (T, t) we conclude that $\mathcal{F}_T$ is pure along X_t as desired. □